Empowering Data-Driven Savings

Standard Operating Procedure

MRO Naming Convention

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PROCEDURE OVERVIEW

The purpose of this document is to systematically generate consistent format and nomenclature for all MRO (Maintenance, Repairs and Operations) materials. These standards apply to all business units using the client specified ERP/EAM system and will be applied to any future additions to the material catalogue.

DEFINITIONS

TERM	DEFINITION
POWER CONVERSIONS	Numbers that are powers of ten can designate values for some characteristics. The most common prefixes are Mega, Kilo, Milli, Micro and K as a suffix. Due to the existing confusion, Milli is abbreviated to M, while Mega and Micro are spelled out. “U” is not normally used because it is not a true representation of Micro.
FRACTIONS VS. DECIMALS	All fractions will be converted to decimals and will use leading zeros but not trailing zeros. (e.g. If .5 then it will read 0.5, if .250 then it will read 0.25, if 1/8 then it will read 0.125)
NUMBER OF ROWS	This information is to be recorded with an unabbreviated word instead of a number. Single row will be spelled out SINGLE ROW, not abbreviated to 1 ROW or SGL ROW. Similarly, double row will be spelled out as DOUBLE ROW, not 2 ROW or DBL ROW.
SPECIFICATIONS	The basic component of a Material value that most often warrants its own characteristics is Specification. The Material Specification values are intended for ANSI, ASTM, or other applicable designations.
NOMINAL SIZE	Size used for the purpose of general identification: the actual size of a part will be approximately the same as the nominal size but need not be exactly the same; for example, a rod may be referred to as 0.25 inch, although the actual dimension of the part is 0.2495 inch, and in this case 0.25 inch is the nominal size.

MATERIAL DESCRIPTION RULES

MATERIAL VALUES

- When material is referenced as a characteristic by itself, the values will be listed as follows:

  PROCESS
  BASE MATERIAL
  SPECIFICATION
  GRADE
  EXAMPLE: GALV STEEL ASTM 105 GRADE B
  EXAMPLE: COLD ROLLED STEEL ASTM 105 GRADE B

  - The exception to the above mentioned arrangement is for Stainless Steels.

    GENERAL ORDER OF STAINLESS STEELS ARE:

    AISI GRADE
    BASE MATERIAL
    SPECIFICATION
    GRADE
    EXAMPLE: 316 SS ASTM 182

- When more than one (1) material is given for a specific item, separate the materials with a slash.

  EXAMPLE: STEEL/NITRILE

- O-RINGS: Use only the material value given in the source description
- Brand names may sometimes be used as a material value when they are a commercial description.

  EXAMPLE: TEFLON or GRAFOIL

SIZE/DIMENSIONS

- Must use a descriptive value (i.e. MM, IN, FT) and specify the dimensional value (i.e. ID, OD, etc.)
- When referencing sizes that are not listed as individual characteristic, list the values in the following sequence: ID, OD, DIA, WD, LG, DP, HT, THK

  Example 0.5IN ID, 0.75IN OD, 3IN LG

  Example 0.5IN ID, 0.75IN OD, 0.25IN THK

    - A space x space (X) would not be used to separate values, except in the case of example below. Specific orders of dimensions should be according to the sample values for each particular noun.
    - Always use a delimiter (e.g. DIA, ID, OD, etc.) after the value when filling in the characteristic of SIZE.
    - If the dimensions are unknown, use space X space (X) between the measurements (e.g. 4 X 6, 1IN X 1.25IN).
- GASKETS are normally sized by ID, OD, THK.

  EXAMPLE: 3IN ID, 4IN OD, 0.125IN THK

  EXAMPLE: 75MM ID, 100MM OD, 6MM THK
- When gaskets have bolt holes use DIA after the bolt hole size.

  EXAMPLE: 0.5IN DIA BOLT HOLE
- O-RINGS are normally sized by the Inside Diameter and Width. An O-RING gauge should be used when determining O-RING sizes.
- NUT SIZES: Use nominal size only. Sometimes called thread size, tube size, pipe size, stud size, screw size, or bolt size. The diameter of a nut is the INSIDE DIAMETER, which is the same size as that of the bolt with which it is to be used.
- STEPPED SHAFT SIZES: When determining the size of stepped shafts, use the overall length only.
- DRILLS can be: Letter sizes (e.g., G, H, or L), Numeric sizes (e.g., 0.5IN, or 0.75IN), Number sizes (e.g., #1, #40, #60)
- Dry Cell Batteries can be letter sizes (e.g., AA, C, D) can also be numeric. If quantities are needed use the (4) AA convention to indicate quantity.
- LARGE, MEDIUM, or SMALL are spelled out when describing sizes
- Quantities should always be enclosed with parenthesis ( ).

  EXAMPLE: (8) 0.5IN DIA BOLT HOLES

TYPE

- If the more specific part is known, the value should indicate its closest level of hierarchy (i.e., a spacer for the inlet vane for the gas turbine, the value for TYPE would be VANE, not the TURBINE). Generally, TYPE excludes brand names or manufacturer types used to describe their products.

ELECTRICAL RATINGS

- GENERAL ORDER OF ELECTRICAL VALUES: VOLTAGE, AMPERAGE, PHASE, HERTZ (CYCLES), WATTAGE, OTHERS (HORSEPOWER, OHM, VOLTAMP, ETC.)

  EXAMPLE: 120VAC 15A 1PH 50/60HZ 6W 75HP

  - EXCEPTION: For Relays, electrical receptacles, and electrical plugs, Amperage is placed before Voltage.
  - When multiple electrical values are used together in the characteristic of **Rating** the values should be separated by using a space (See example above).
- A slash (/) should be used for separation of values when referring to more than one voltage rating.

  EXAMPLE: 240 and 480VAC = 240/480VAC
- A (TO) will be used to show all ranges. EXAMPLE: 4 TO 12 AMPS = 4 TO 12A. Do not use dash (-) to signify range.
- When VAC and VDC are used together, always put the VAC first and use a comma to separate the two.

  EXAMPLE: 120VAC and 20VDC = 120VAC, 20VDC
  380VAC and 380VDC = 380VAC/DC
- AC should be added to the voltage when Hertz (HZ) or Phase (PH) is given.
- When Volts or Amps are over 1,000 convert to Kilo.

  EXAMPLE: 1,000V = 1KV
  2,500V = 2.5KV
  40,000A = 40KA
- When voltages of opposite polarities (positive/negative) are listed in the same characteristic value, use the following as an example.

  EXAMPLE: CORRECT = +/-15VDC INCORRECT = +15VDC-15VDC

THREADS/CONNECTION

- GENERAL ORDER OF THREAD DESIGNATIONS ARE:

  NUMBER OF THREADS PER INCH (PITCHES PER INCH), THREAD SERIES SYMBOL (e.g. UNC, UNRC, UNF, UNRF, UNEF, UNREF), THREAD CLASS SYMBOL (CLASS FIT), HAND (RIGHT-HAND or LEFT-HAND)

  EXAMPLE: 20 UNC-2A LH

  20 = NUMBER OF THREADS PER INCH (PITCHES PER INCH)

  UNC = THREAD SERIES SYMBOL (UNIFIED COARSE THREAD)

  2A = THREAD CLASS SYMBOL

  LEFT HAND = LH

  RIGHT HAND = RH

- Use “Compression, Flare, etc.” as the connection value when describing a tube connection with a nut and/or ferrule.

- When referring to connection threads, they should be listed in the following sequence (when connection is not a separate characteristic): SIZE, MALE or FEMALE, INDUSTRY CONNECTION TYPE.

  EXAMPLE: 1.5IN MNPT

- Make sure the sizes associate with the correct connection.

  EXAMPLE: 0.5IN MNPT, 1IN FNPT

  SIZE = 0.5IN, 1IN

  CONNECTION = MNPT, FNPT

- If size and connection are separate characteristics, connection threads should be listed as MALE or FEMALE, INDUSTRY CONNECTION TYPE.

  EXAMPLE: MNPT, FNPT

- **DO NOT** change NPTF to FNPT. When connection indicates NPTF, the F does not mean “female” (NPTF is a Dry seal thread).

PRESSURE RATINGS

- Convert Class Ratings (e.g. weld flanges) and PSI Ratings to LB (does not apply to tubing or other specialty products.)

  EXAMPLE: CORRECT = 150LB / INCORRECT = Class 150 or 150PSI

  - PSI is valid for gauges and set pressure ratings on relief valves.
  - DO NOT convert PSIG or PSIA Ratings.

- Use the design pressure rating at a predetermined temperature (if available) in characteristic of Temperature Rating on valves.

  EXAMPLE: 1600LB @ 600DEG F

- Example of Standardization Rating:

Source	Standardization
0-30 #	0 TO 30PSI
0-30 PRSR	0 TO 30PSI
0-30 LBS PSIG	0 TO 30PSI
0-30 PSIG	0 TO 30PSI

- Pipe Schedules may sometimes be used as a rating value (e.g. SCH 40, SCH 80, etc.)

VALUE PRESENTATION & ABBREVIATIONS

- VALUE SNUGGLED AGAINST ABBREVIATION: The following are values that should always be abbreviated with the number snuggled against the abbreviation.

  EXAMPLE: 20 AMPS would be typed as: 20A

VALUE	**ABBREVIATION USED**
VOLTS V	V
VOLTS AC	VAC
VOLTS DC	VDC
AMPS	A
HERTZ OR CYCLES PER SECOND	HZ
PHASE	PH
MEGAWATTS	MW
MILLION VOLT AMP	MVA
KILOVOLT	KV
VOLT AMP	VA
WATTS	W
KILOVOLT AMP	KVA

KILOVOLT AMP REACTIVE	KVAR
PERCENTAGE	PCT
MILLIMETERS	MM
METERS	M
POUNDS	LB
POUND-FORCE PER SQUARE INCH	PSI
HORSE POWER	HP
OHMS	OHM
DEGREES FAHRENHEIT	DEG F
DEGREES CELSIUS	DEG C
REVOLUTIONS PER MINUTE	RPM

- Abbreviations should be used only where they're meaning is unquestionably clear. **WHEN IN DOUBT, SPELL IT OUT.**
- NOTE: VALUE PRESENTATION SHOULD ALWAYS BE IN ACCORDANCE WITH THE RESPECTIVE NOUN/MODIFIER PAIR SAMPLE VALUES AND CHARACTERISTIC COMMENTS.
- **DO NOT** use a period at the end of an abbreviation.

MANUFACTURER PART NUMBER

- Capture the Manufacturer Part Number just as the Manufacturer publishes it.
- Convert all Dodge 6-digit numbers to the alpha-numeric nomenclature. (124170 to F2B-SC-012)